Long-Range Underwater Acoustic Channel Estimation

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***Abstract*— Long-range underwater acoustic communication (LR-UWAC) is an essential technique for applications such as the control of unmanned underwater vehicles, gliders, or tactical submarines for long-term monitoring tasks. While techniques for estimating the channel impulse response (CIR) of short-range UWAC have been successfully applied, this is not the case for LR-UWAC. Here, the main challenge is to handle the diverse channel structure, which can be of a sparse or a non-sparse structure. In this paper, we propose a robust channel estimator for LR-UWAC. Our solution includes a switching mechanism to classify the channel as either sparse or non-sparse. In its sparse form, the LR-UWAC channel is characterized by a long time delay spread, time-invariant characteristics, and a block structure. Thus, we propose a non-trivial combination of the block subspace pursuit and the distributed subspace pursuit algorithms to exploit the channel's block structure. For non-sparse LR-UWAC channel structures, we approximate the received signal as Gaussian, and derive a maximum-likelihood (ML) estimator to separate the non-resolvable multipath. Extensive numerical simulation and results from two long-range sea experiments demonstrate the efficiency of our approach in identifying the channel's structure, and to accurately estimate the channel compared to state-of-the-art benchmarks.**

***Index Terms*— Long range underwater acoustic communication, diverse channel structure, block structure, non-resolvable multipath, channel estimation.**

I. Introduction

WITH the advance in underwater acoustic communications, applications such as seabed mapping, ocean current monitoring, and search and survey military missions have now become practical. Such long-term monitoring tasks are mostly handled by autonomous underwater vehicles (AUVs), gliders, or tactical submarines. While these vehicles can travel over long distances, the control over such vehicles requires robust long-range underwater acoustic communication (LR-UWAC). The considered scenario involves a one-way transmission to a remote node, mostly to change its current mission [1], [2], [3].

The absorption coefficient in water increases with the carrier frequency. Thus, for long distance underwater acoustic communication, we adopt a low frequency transducer for low acoustic absorption. Here, compared to the short distance underwater acoustic communication, long distance underwater acoustic communication has more diverse CIR, which can be both simple and complex. Specifically, while the CIR of short-range UWAC are naturally sparse [4], [5], [6], the CIR of LR-UWAC can be both sparse or non-sparse with a delay that can sometimes spread over more than a second [7], [8], [9]. In turn, the channel structure is diverse with a mixture of sparse and non-sparse components [1]. On the other hand, compared to the short-range channel, whose coherence time is on the order of a few tens of milliseconds, the LR-UWAC's CIR is relatively time-invariant [2], [9]. Fig. 1 presents two examples of LR-UWAC channels, which we recorded across the shores of Haifa, Israel, at distances of 90 km in different depths. We observe a diverse channel structure and a time delay spread of around 300 ms. The long time delay spread of LR-UWAC channels results in serious inter-symbol interference (ISI), and the expected diversity in the channel's structures calls for the design of a robust/adaptive receiver for LR-UWAC.

In this work, we describe a robust channel estimator that is specifically tailored to LR-UWAC and can be applied to both sparse and non-sparse structures. Channel estimation for sparse or non-sparse LR-UWAC has been explored in the literature. In the framework of compressed sensing, sparse channel estimation exploits the diversity within the channel. Examples are orthogonal matching pursuit (OMP) [10], subspace pursuit (SP) [11], [12], [13], and smoothed l_0-norm (SL0) [14]. When the channel's structure is non-sparse, least square (LS) [15], [16], and adaptive algorithm [17], [18], [19], [20] can adjust the resulting coefficients in the search for an optimal solution. While the current approaches for LR-UWAC have also demonstrated the capability to communicate well at rates of 400 bits per second [1] in sea trials, there is still the challenge of ensuring robust communication.

Given the differences between the receiver's structures for sparse and non-sparse channels, immediately following the initial channel estimation, we determine the channel's type and accordingly perform sparse or non-sparse channel estimation. For sparse channels, based on the block subspace pursuit (BSP) algorithm [12] and the distributed subspace pursuit (DSP) algorithm [13], we propose a non-trivial combined

Manuscript received 25 May 2022; revised 29 November 2022; accepted 29 January 2023. Date of publication 8 February 2023; date of current version 12 September 2023. This work was supported in part by the North Atlantic Treaty Organization (NATO) Science for Peace and Security Program (SAFE-UComm) under Grant G5884. An earlier version of this paper (channel estimation for the sparse channel) was presented in part at the 2022 Underwater Communications and Networking Conference [DOI: 10.1109/UComms56954.2022.9905683]. The associate editor coordinating the review of this article and approving it for publication was L. Simic. *(Corresponding author: Roee Diamant.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TWC.2023.3241230.

Digital Object Identifier 10.1109/TWC.2023.3241230

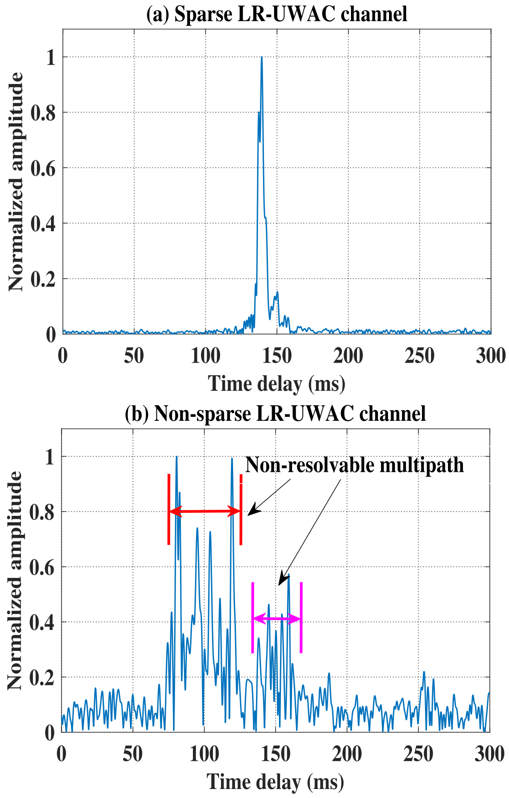


Fig. 1. Different sparse structures of the LR-UWAC channel. (a) Sparse LR-UWAC channel with a receiver depth of 300 m. (b) Non-sparse LR-UWAC channel with a receiver depth of 100 m.

solution referred to as the simultaneous block subspace pursuit (SBSP) algorithm.

For non-sparse channel structures, we formulate a new maximum-likelihood (ML) estimator that is able to separate the non-resolvable multipath. Specifically, for the non-sparse LR-UWAC channel with a non-resolvable and dense multipath, we approximate the received signal as Gaussian, and follow normal distribution [21], [22], [23], [24]. Then, we drive an ML channel estimator for the non-sparse LR-UWAC channel.

The contribution of our work is threefold:

1) A non-trivial combination of the DSP and BSP algorithms for LR-UWAC sparse channel estimation.
2) A novel ML channel estimator for LR-UWAC for non-sparse channels with a non-resolvable and dense multipath.
3) A switching mechanism to differentiate between sparse and non-sparse channel estimators.

We evaluate our method's performance in three ways: by numerical simulations; using previously recorded signals from a LR-UWAC sea experiment; and in another designated LR-UWAC sea experiment. The results show that the LR-UWAC channel discriminates well between sparse and non-sparse channels. In terms of communication capabilities, better performance is obtained compared to popular sparse channel estimation algorithms, namely, the SL0 [14], the OMP [10], the BSP [12], and the DSP [13]; and compared to non-sparse channel estimation, namely, the LS [16], the recursive least squares (RLS) [17], and the hybrid [25], [26], [27] channel estimator, as well as the well known MUSIC algorithm [6].

The remainder of this paper is organized as follows: In Section II, we describe the state-of-the-art for UWA channel estimation as well as methods for LR-UWAC. Section III presents our system model and main assumptions. Section IV describes the proposed channel estimation scheme. Results from modeled numerical simulations and from two designated sea experiments are presented in Section V. Conclusions are drawn in Section VI.

Notation: Bold capital letters denote matrices. The superscripts $*$, T, and H denote the conjugate, transpose, and conjugate transpose operator, respectively. The set operations $\cup$ denotes the union operation. $|\mathbf{x}|$ denotes the absolute value of the vector $\mathbf{x}$. $||\mathbf{x}||_2$ and $||\mathbf{x}||_1$ represent the l_2 norm (Euclidean norm) and l_1 norm of the vector $\mathbf{x}$, respectively. $|\text{supp}(\mathbf{a})|$ denotes the number of nonzero elements of the vector $\mathbf{a}$, i.e., $||\mathbf{a}||_0 = |\text{supp}(\mathbf{a})|$. $\mathbf{A}^\dagger$ represents the matrix pseudo inverse. $\mathbf{Re}\{\bullet\}$ denotes the real part of a complex-valued expression. $\odot$ denotes the component-wise (Schur) product. diag$\{\mathbf{x}\}$ denotes a square diagonal matrix with the elements of vector $\mathbf{x}$ on the main diagonal.The $M \times M$ discrete Fourier transform (DFT) matrix is $\mathbf{F}_{m,n} = 1/\sqrt{M}[e^{-j\frac{2\pi}{M}mn}], m, n \in 0, 1, \cdots, M-1$ (where $j = \sqrt{-1}$).

II. Related Work

Over the last decade, several solutions have been proposed for LR-UWAC. The authors of [2] and [28] explored the chirp spread spectrum (CSS) modulation and the double-differentially coded spread-spectrum system for LR-UWAC, respectively. Combined with beamforming, a quadrature-phase shift-keying (QPSK) modulation was proposed and achieved good results in a LR-UWAC experiment [29]. Still, the modulation seems to fit a specific channel type. For this reason, based on environmental information and by training on various channel types, an adaptive approach was proposed to pre-set the modulation scheme according to a prediction of the LR-UWAC CIR [1].

Different receivers were proposed for LR-UWAC. A modal dispersion compensation receiver is proposed in [3], where a waveform expansion of high-order modes is compensated. To increase the spectral efficiency, [9] reported a LR-UWAC experiment using a high-order constellation and passive time reversal combined with a decision-feedback equalizer (DFE). A time reversal receiver [30] was proposed for LR-UWAC vertical sensor array processing, which can effectively mitigate the ISI and reduce the bit error rate (BER). On top of these, channel estimation algorithms are also applied for LR-UWAC. The OMP channel estimator is modified for multi-carrier orthogonal frequency division modulation (OFDM) receivers in LR-UWAC using a towed horizontal array [8]. Alternatively, combining three transmissions to decode an 8-phase shift-keying communication, [31] proposed an OMP channel-estimation-based equalizer for the LR-UWAC. However, while tried in field, the above solutions considered the LR-UWAC channel as perfectly sparse or non-sparse, and no solution is given for mixed cases.

Adaptive channel estimation algorithms were developed for non-sparse channel estimation, where adaptivity is employed over modulation schemes [18], transmission rates [19], channel coding schemes [20], and interference cancellation [17]. These adaptive algorithms can maximize communication performance for different system parameters. Existing solutions describe a hybrid sparse/diffuse model [25], [26], [27] for non-sparse channel estimation. By modeling the channel with both diffuse and sparse components, a hybrid sparse/diffuse model is proposed to estimate the UWA channel. However, this algorithm simplifies the usage of the Rayleigh fading assumption for the diffuse component. For the non-sparse LR-UWAC channel with non-resolvable and dense multipath, the diffuse component may not follow Rayleigh fading, while in [32] the Ricean fading channel model is shown to be more suitable than the Rayleigh fading channel model. Thus, a hybrid sparse/diffuse model would not fully exploit the non-resolvable multipath and may not perform well due to high uncertainty in the structure of the LR-UWAC channel. To the best of our knowledge, no adaptive solution has been offered to fit the case of either a sparse or a non-sparse LR-UWAC channel.

When the UWA channel is sparse, its diversity can be exploited for performance enhancement by means of compressed sensing. For cases of low signal-to-noise ratio (SNR), subspace pursuit-based algorithms have been proposed for channel estimation [11], [12], [13]. Designed for channels with a cluster-like pattern, the structured approximation l_0 (SAL0) norm algorithm is proposed to exploit the block structure of the UWA channel [33]. While achieving an accurate reconstruction of the channel, the method is sensitive to low SNR and to different noise distribution types. Furthermore, the above compressed sensing approaches are more suitable for short-range UWAC and may not manage the complexity of the LR-UWAC channel. When array processing is available, high resolution in separation of the channel's taps is possible. This is exploited by the MUSIC [6] and ESPRIT algorithms [34], linked to the challenge of channel estimation. Another high-resolution solution is found in the SPICE algorithm [35], [36]. However, we report that testing SPICE for estimating LR-UWAC CIR fails when the channel's taps are too dense.

In this work, we focus on two remaining challenges for estimating the LR-UWAC CIR. First, the diversity embedded within the LR-UWAC channel that yields either sparse or non-sparse structures should be considered. Second, to the best of our knowledge, no previous work has considered a high-resolution channel estimation for the LR-UWAC channel when the multipaths are non-resolvable. The developed channel estimators are designed for the complex environment of the long channel impulse response. However, the solutions developed can also apply to environments rich with multipath such as harbors or reef environments.

III. System Model

A. Main Assumptions

Our system setup comprises a transmitter emitting low-frequency acoustic communication signals; and a receiver, deployed at distances on the order of 100 km, aiming to decode the communication. Regarding our sea trial, we focus on the frequency range of 800 Hz to 1500 Hz and transmissions at a source level of roughly 200 dB re 1μPa at 1 m. At the 100 km transmission range, the SNR is expected to be on the order of 20 dB. The transmitter is assumed to be fixed, e.g., deployed from a buoy or a boat, while the receiver may move, e.g., an AUV, glider, or drifter. In all cases, we assume a slow time-varying channel model. An illustration of the bathymetry profile and the historical sound speed profile in Haifa Bay, Israel is presented in Fig. 2, and is further discussed in more detail below in Section VI-A.

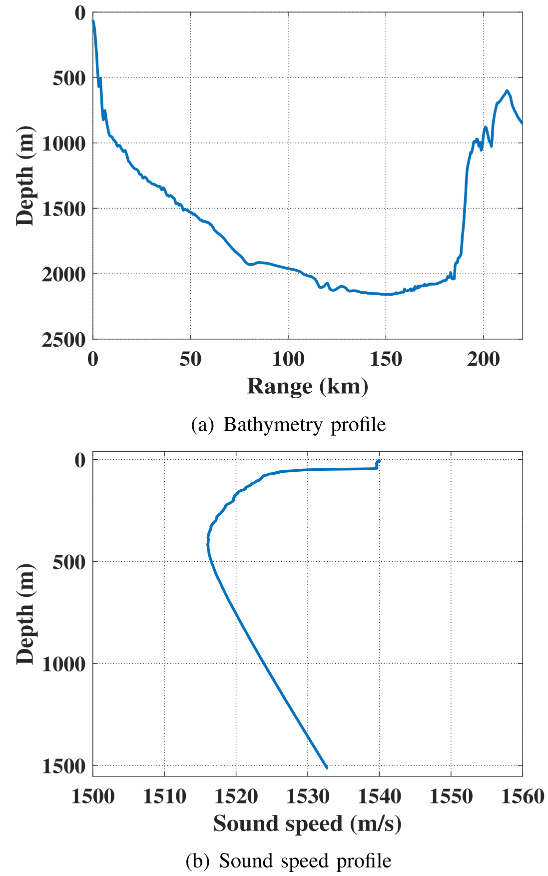


Fig. 2. Bathymetry profile and sound speed profile. (a) Bathymetry profile from the transmitter to the receiver. (b) Sound speed profile.

B. Problem Definition

Our goal in this work is to estimate the LR-UWAC channel. We focus on an OFDM modulation that was previously proposed for LR-UWAC [8]. In an OFDM system, the source data are modulated by inverse DFT (IDFT) on parallel subcarriers. The transmitted OFDM symbol $x(t)$ in the time domain is

$$x(t) = \frac{1}{\sqrt{M}} \sum_{m=0}^{M-1} \chi(m) e^{j\frac{2\pi}{M}tm}, \quad 0 \leq t \leq K-1, \tag{1}$$

where $\chi(m)$ are the OFDM symbols at subcarrier m, $0 \leq m \leq M-1$, and $\mathbf{X} = [\chi(0), \chi(1), \cdots, \chi(M-1)]^T$. We assume $\chi(m)$ is a zero mean random variable with $|\chi(m)|^2 = 1$.

Observing the diverse channel structure from the examples in Fig. 1, we characterize the channel as comprising of two possible configurations for the channel's formulation: a sparse and a non-sparse channel. We assume that the channel between the transmitter and the receiver is a linear time-invariant (LTI) finite impulse response (FIR) filter. Formally, the LR-UWAC channel can be expressed by

$$\mathbf{h}_t = \sum_{n=0}^{N-1} h(n)\delta(t-\tau(n)), \tag{2}$$

where N is the path number of the LR-UWAC channel, $h(n)$ is the amplitude of n-th multipath, $\tau(n)$ is the delay associated with the n-th multipath, and $\delta(t)$ is the Dirac delta. After removing the guard interval and performing DFT demodulation, the received OFDM signal can be expressed by

$$y(m) = \chi(m)c(m) + w(m), \quad 0 \le m \le M-1, \tag{3}$$

where $w(m)$ represents additive white Gaussian noise with mean 0 and variance σ_w^2, and $c(m)$ is the Fourier transform of the CIR at subcarrier m, namely,

$$c(m) = \frac{1}{\sqrt{M}} \sum_{n=0}^{N-1} h(n) e^{-j\frac{2\pi}{M}mn}. \tag{4}$$

Hence, we have

$$\mathbf{H} = [c(0), c(1), \cdots, c(M-1)]^T = \mathbf{F}_N \mathbf{h}, \tag{5}$$

where $\mathbf{h} = [h(0), \cdots, h(N-1)]^T$ is the real-valued LR-UWAC CIR and $\mathbf{F}_N$ is a $M \times N$ submatrix of the DFT matrix $\mathbf{F}$. By matrix representation,

$$\mathbf{Y} = \mathbf{X}_D \mathbf{F}_N \mathbf{h} + \mathbf{W}, \tag{6}$$

where

$$\mathbf{X}_D = \mathrm{diag}\{\chi(0), \chi(1), \cdots, \chi(M-1)\}, \tag{7a}$$
$$\mathbf{Y} = [y(0), y(1), \cdots, y(M-1)]^T, \tag{7b}$$
$$\mathbf{W} = [w(0), w(1), \cdots, w(M-1)]^T. \tag{7c}$$

1) System Model for Sparse Channel Estimation: In sparse channels, (6) can be rewritten as

$$\mathbf{Y} = \mathbf{S}\mathbf{h} + \mathbf{W}, \tag{8}$$

where

$$\mathbf{S} = \mathbf{X}_D \mathbf{F}_N \tag{9}$$

is the measurement matrix. Therefore, the problem of sparse LR-UWAC channel estimation can be formalized by

$$\arg\min ||\mathbf{h}||_0 \quad s.t. \quad ||\mathbf{Y} - \mathbf{S}\mathbf{h}||_2^2 \le \varepsilon, \tag{10}$$

where ε is the noise factor. Thus, we can convert the estimation of the LR-UWAC channel into a search for the solution of (10), and apply compressed sensing techniques for channel estimation.

2) System Model for Non-Sparse Channel Estimation: For the case of a non-sparse LR-UWAC channel, we derive the ML channel estimator,

$$\tilde{\mathbf{h}} = \arg\max_{\mathbf{h}} f(\mathbf{Y}; \mathbf{h}), \tag{11}$$

where f is the probability density function, $\mathbf{Y} \in \mathbb{C}^{M \times 1}$ is the received LR-UWAC signal, and $\mathbf{h}$ is the LR-UWAC channel. Since the multipath in the non-sparse LR-UWAC case is dense, the received signal is a sum of a large number of independent random variables. Thus, according to the central limit theorem, the received signal can be modeled as Gaussian, and follows a normal distribution [21], [22], [23], [24].

IV. Derivation of the Proposed Algorithm

Our solution for the LR-UWAC channel estimation is illustrated in the block diagram in Fig. 3. We start with a coarse channel estimation to set our system's parameters and to determine whether the current channel is sparse or non-sparse. The former is performed using a least square approach over a training sequence, while for the latter we use the Gini index to quantify the channel's *sparsity* [37], [38]. Based on this decision, we continue with either sparse estimation or non-sparse estimation. The former involves the SBSP algorithm, while the latter is based on an ML estimator. In the following, we discuss the details of each of the above components.

A. Coarse Channel Estimation

The coarse LR-UWAC channel estimator is based on the least square approach. Formally, the LS estimation [15], [16] is given by

$$\mathbf{h}^{LS} = (\mathbf{S}^H\mathbf{S})^{-1}\mathbf{S}^H\mathbf{Y}, \tag{12}$$

where $\mathbf{Y}$ and $\mathbf{S}$ are the received OFDM signal and the measurement matrix, respectively. Compared to the CS algorithm, the LS method is easy to implement, and has a lower computational complexity. Then, based on the result in (12), the LR-UWAC channel is classified as either sparse or non-sparse.

B. Sparse Detection

Metrics to measure the channel's sparsity have been presented in the literature [37], [38]. Possible metrics are the number of multipath components (MPCs), the channel's degrees of freedom (DoF), the Gini index, and the Ricean K factor. Among these, we chose the Gini index. This is because, while the number of MPCs intuitively indicates the change of channel sparsity, the MPCs with low amplitude would have only a small impact on channel sparsity. Furthermore, it is difficult to separate the line-of-sight (LOS) and the non-LOS components for the Ricean K factor. The channel's DoF and Gini index can provide good performance [37], but the former has higher computational complexity, since it involves calculating the rank of the channel's correlation matrix. In turn, the Gini index satisfies Dalton's 1st Law [39] that a sparse measure would decrease if the energy of an element spreads over other elements. The Gini index is also scale-invariant and independent of the channels' total energy [37].

Following the coarse channel estimation in (12), the Gini index is defined by [37], [38]

$$\Psi = 1 - 2\sum_{n=0}^{N-1} (\frac{h(n)}{\|\mathbf{h}\|_1} . \frac{N-n-0.5}{N}), \tag{13}$$

where $\mathbf{h} = [h(0), h(1), \ldots, h(N-1)]$ is the real-valued amplitude of the channel's components that are arranged

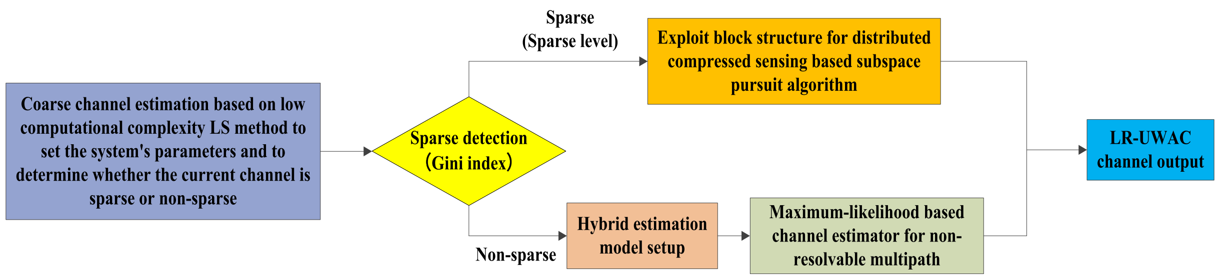


Fig. 3. The flow chart of LR-UWAC channel estimation.

from small to large, and N is the length of the LR-UWAC channel. For a non-sparse channel whose components hold equal amplitude, we have $\Psi = 0$, whereas for a highly sparse channel with a single dominant component, we have $\Psi = 1$.

Thresholding ψ, we determine whether the channel is sparse or not. We propose the following procedure to set the threshold: Based on the parabolic equation (PE) model [40], we generate a large set of CIRs from different bathymetry, sound speed profile, sea surface roughness, and bottom sediment composition conditions and for different transmitter and receiver depth setups, and calculate their Gini index. The test channels have been initially classified to 'sparse' and 'non-sparse' by the assumption that when the depths of the receiver and the transmitter are 10 m lower than the first dominant bathymetric layer in the sound speed profile, the simulated channel is referred to as sparse, and vice versa for a non-sparse channel. These indexes are then treated as random variables under a mixed distribution model with two classes, namely, sparse and non-sparse. Differentiating between the two can be achieved by applying the expectation maximization algorithm [41] or, in the special case of Gaussian mixture, the Baum-Welch algorithm [42]. The result is an association of each CIR to either class '1' or class '2' as sparse channel and non-sparse channel, respectively. We then numerically calculate the probability density function (PDF) of the two classes and determine the threshold to be the one that yields the best precision-recall tradeoff.

The kernel density estimation method is applied to estimate the PDF by [43]

$$\tilde{f}(x) = \frac{1}{Lb}\sum_{i=1}^{L} K\left(\frac{x - g(i)}{b}\right), \tag{14}$$

where $K(\cdot)$ is the kernel smoothing function, b is the bandwidth, L is the number of the Gini index values of the CIRs, and $g(i)$ is the Gini index value. The estimate is based on a normal kernel smoothing function, which is defined by

$$K(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}. \tag{15}$$

The optimal smoothing bandwidth parameter b for the Gaussian kernel density estimation is given by [44]

$$b = 1.06\varpi L^{-\frac{1}{5}}, \tag{16}$$

where ϖ is the standard deviation of the Gini index values. To maximize the possibility that each channel is well classified to a channel type, we select the cross point of two PDF curves for the sparse and the non-sparse channels as the detection threshold. An example of such a tradeoff from our numerical investigation is presented further below in Fig. 7. Next, we present the details of the sparse and non-sparse channel estimation.

C. Sparse LR-UWAC Channel Estimation

In its sparse form, the LR-UWAC channel is characterized by a long time delay spread, time-invariant characteristics, and a block structure. The latter refers to a cluster-like pattern of the channel's taps. Considering these characterizations, we perform channel estimation by combining the block subspace pursuit (BSP) algorithm and the distributed subspace pursuit (DSP) algorithm. The former's aim is to exploit the block structure of the LR-UWAC channel, and the latter derives a joint sparse recovery algorithm for efficient LR-UWAC channel estimation. Specifically, the DSP algorithm treats the multipath arrivals as sparse solutions with a common time support, which is adopted to derive a joint sparse recovery algorithm for efficient channel estimation. The result is a decrease in the observation time window needed to recover the channel's multipath structure. Combining BSP and DSP, our SBSP scheme treats the channel as a concatenation of multiple blocks, thereby utilizing the temporal correlations that exist - not per channel, but per such blocks of taps in the channel. Still, simply concatenating the BSP and the DSP algorithms is not possible, since the former is used over one observation window while the latter uses multiple such windows. Thus, the SBSP is far from being a simple integration procedure.

1) Problem Formulation: In this study, the blocks of the sparse LR-UWAC channel are assumed contiguous. Motivated by l_2/l_0-norm algorithm for recovering block-sparse signals [45], [46], we consider the LR-UWAC channel as a concatenation of η blocks with a same block size d. Thus, the block-sparse channel $\mathbf{h}_t$ can be expressed as

$$\begin{aligned}\mathbf{h}_t = [&\underbrace{h_t(0),\cdots,h_t(d-1)}_{\mathbf{h}_t^T[1]},\underbrace{h_t(d),\cdots,h_t(2d-1)}_{\mathbf{h}_t^T[2]},\\ &\cdots,\underbrace{h_t(N-d),\cdots,h_t(N-1)}_{\mathbf{h}_t^T[\eta]}]^T,\end{aligned} \tag{17}$$

where $\mathbf{h}_t \in \mathbb{C}^{N\times 1}, 1 \leq t \leq l$ denotes the LR-UWAC channel of some t-th continuous measurement; $\mathbf{h}_t[j], 1 \leq t \leq l, 1 \leq j \leq \eta$ denotes the j-th sub-block channel of

t-th continuous measurement, note that $N = \eta d$. While the BSP method achieved similar performance for different block sizes [12], the larger block size d would better exploit the channel's intra-block correlation, which is arguably the case for the LR-UWAC channel with less sparsity. Thus, we set a larger d for a channel with a smaller Gini index. A threshold 0.71 for Gini index is set to the sparse detection according to the simulation results in Section V-B. Below this threshold, we consider the detected channel as a non-sparse channel. Above this threshold we divide the Gini Index in range (0.71 1] into regions and block sizes by the rule that a larger block size would fit a channel corresponding to a smaller Gini index. Specifically, for the intervals (0.95,1], (0.90,0.95], (0.85,0.90], and (0.71,0.85] of the Gini index, we set $d = [2, 4, 6, \text{and } 8]$, respectively. We define the number of the nonzero blocks κ of the channel $\mathbf{h}_t$ as

$$\kappa = ||\mathbf{h}_t||_{2,0} = \sum_{j=1}^{\eta} I(||\mathbf{h}_t[j]||_2 > 0) \tag{18}$$

with the indicator function $I(\cdot)$. Thus, the estimation of a sparse channel can be expressed by the optimization problem

$$\arg\min_{\mathbf{h}} \sum_{t=1}^{l} (||\mathbf{h}_t||_{2,0}) \quad s.t. \quad ||\mathbf{Y} - \mathbf{S}\mathbf{h}||_2^2 \le \varepsilon, \tag{19}$$

where ε is the noise factor, and

$$\mathbf{S} \in \mathbb{C}^{Ml \times Nl} = \mathrm{diag}\{\mathbf{S}_1, \mathbf{S}_2, \cdots, \mathbf{S}_l\}, \tag{20a}$$
$$\mathbf{h} \in \mathbb{C}^{Nl \times 1} = [\mathbf{h}_1, \mathbf{h}_2, \cdots, \mathbf{h}_l]^T, \tag{20b}$$
$$\mathbf{Y} \in \mathbb{C}^{Ml \times 1} = [\mathbf{Y}_1, \mathbf{Y}_2, \cdots, \mathbf{Y}_l]^T. \tag{20c}$$

$\mathbf{S}_t \in \mathbb{C}^{M \times N}, 1 \le t \le l$ denotes t-th continuous measurement, $\mathbf{S}_t[j] \in \mathbb{C}^{M \times d}$ consists of the columns of $\mathbf{S}_t$ with indices $(j-1)d + 1$ to jd for $1 \le j \le \eta$. $\mathbf{Y}_t \in \mathbb{C}^{M \times 1}, 1 \le t \le l$ denotes t-th data block of the received signal. By combining continuous measurements with channel blocks, we solve (19) to jointly recover sparse blocks of taps in the channel as follows:

2) The SBSP Algorithm: The SBSP algorithm works iteratively. In each iteration, we take atom $\mathbf{S}_t, t = 1, \cdots, l$ from $\mathbf{S}$, and compute the inner product of $\mathbf{S}_t$ with the residual error $\mathbf{Y}_{r,t}$. Then, we sum up the inner product outputs of l continuous measurements, and select the κ first block indices in descending order, as the one that obtains the highest correlation with the residual error $\mathbf{Y}_{r,t}$. Formally,

$$\mathbf{z}_t^i = \mathbf{S}_t^H \mathbf{Y}_{r,t}^{i-1}, \ t = 1, \cdots, l, \tag{21a}$$
$$\tilde{T}^i = T^{i-1} \cup \{\kappa \text{ first block indices in descending sort} \sum_{t=1}^{l} \|\mathbf{z}_t^i[j]\|_2, \ 1 \le j \le \eta\}, \tag{21b}$$

where $\mathbf{z}_t \in \mathbb{C}^{N \times 1} = [\mathbf{z}_t[1], \mathbf{z}_t[2], \cdots, \mathbf{z}_t[\eta]]$, i is the iteration number, T^i denotes the selected κ block sets at the i-th iteration.

According to the selected κ block sets, we calculate the coefficients of the LR-UWAC channel by means of least squares. Note that the κ first blocks correspond to the largest magnitude blocks in the vector $\mathbf{h}_t^i$. Hence,

$$\mathbf{h}_t^i = \mathbf{S}_{\tilde{T}^i,t}^{\dagger} \mathbf{Y}_t, \ t = 1, \cdots, l, \tag{22a}$$
$$T^i = \{\kappa \text{ first block indices in descending sort} \sum_{t=1}^{l} \|\mathbf{h}_t^i[j]\|_2, \ 1 \le j \le \eta\}. \tag{22b}$$

Then, the residual vector is calculated by

$$\mathbf{Y}_{r,t}^i = \mathbf{Y}_t - \mathbf{S}_{T^i,t} \mathbf{h}_{T^i,t}^i, \ t = 1, \cdots, l. \tag{23}$$

The above steps are repeated until the stopping criterion

$$\min\left\|\mathbf{Y}_{r,t}^i\right\|_2 \ge \min\left\|\mathbf{Y}_{r,t}^{i-1}\right\|_2, \ t = 1, \cdots, l, \tag{24}$$

is satisfied. At the final iteration, the channel's estimate $\tilde{\mathbf{h}}_t$ is obtained by

$$\tilde{\mathbf{h}}_{\{[0,\cdots,N-1]-T^i\},t} = \mathbf{0}, \tag{25a}$$
$$\tilde{\mathbf{h}}_{T^i,t} = \mathbf{S}_{T^i,t}^{\dagger} \mathbf{Y}_t, \ t = 1, \cdots, l. \tag{25b}$$

Table I summarizes the main steps of the SBSP algorithm. Note that when $l = 1$, the SBSP algorithm converges to the BSP algorithm, and when $d = 1$, the SBSP algorithm converges to the DSP algorithm.

TABLE I
SIMULTANEOUS BLOCK SUBSPACE PURSUIT ALGORITHM (SBSP)

Input : measurement matrix $\mathbf{x}$, continuous measurements number l, received signal $\mathbf{y}$, nonzero block number κ_t, block size d.

Initialize :
  $i = 0$;
  $\mathbf{z}_t^i = \mathbf{S}_t^H \mathbf{Y}_t, t = 1, \cdots, l$;
  $T^0 = \{\kappa_t$ first block indices in descending sort $\sum_{t=1}^{l} \|\mathbf{z}_t^i[j]\|_2, \ 1 \le j \le \eta\}$;
  $\mathbf{Y}_{r,t}^i = \mathbf{Y}_t - \mathbf{S}_{T^0,t}^i \mathbf{z}_{T^0,t}^i, t = 1, \cdots, l.$
Iteration :
  $i = i + 1$;
  $\mathbf{z}_t^i = \mathbf{S}_t^H \mathbf{Y}_{r,t}^{i-1}, t = 1, \cdots, l$;
  $\tilde{T}^i = T^{i-1} \cup \{\kappa_t$ first block indices in descending sort $\sum_{t=1}^{l} \|\mathbf{z}_t^i[j]\|_2, \ 1 \le j \le \eta\}$;
  $\mathbf{h}_t^i = \mathbf{S}_{\tilde{T}^i,t}^{\dagger} \mathbf{Y}_t, t = 1, \cdots, l$;
  $T^i = \{\kappa_t$ first block indices in descending sort $\sum_{t=1}^{l} \|\mathbf{h}_t^i[j]\|_2, \ 1 \le j \le \eta\}$;
  $\mathbf{Y}_{r,t}^i = \mathbf{Y}_t - \mathbf{S}_{T^i,t} \mathbf{h}_{T^i,t}^i, t = 1, \cdots, l$;
  If $\min\left\|\mathbf{Y}_{r,t}^i\right\|_2 \ge \min\left\|\mathbf{Y}_{r,t}^{i-1}\right\|_2, t = 1, \cdots, l,$
  then $T^i = T^{i-1}$ and quit.
Output :
  The estimate channel $\tilde{\mathbf{h}}_t$, as $\tilde{\mathbf{h}}_{\{[0,\cdots,N-1]-T^i\},t} = \mathbf{0}$,
  and $\tilde{\mathbf{h}}_{T^i,t} = \mathbf{S}_{T^i,t}^{\dagger} \mathbf{Y}_t, t = 1, \cdots, l.$

3) The Convergence Analysis of the SBSP Algorithm: The convergence of the SBSP algorithm is analyzed in this numerical example. We are primarily interested in the relative error [47]

$$\mathrm{RE} = \frac{||\tilde{\mathbf{h}}^i - \mathbf{h}||_2^2}{||\mathbf{h}||_2^2} \tag{26}$$

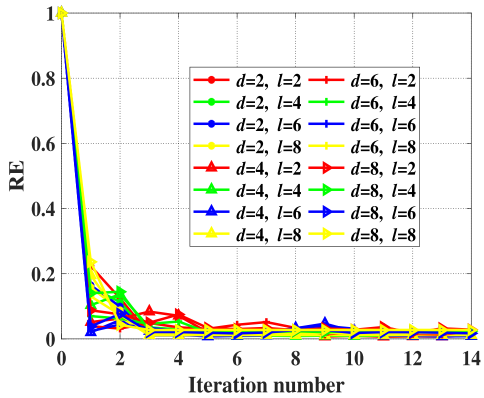


Fig. 4. The convergence of the SBSP algorithm for a different block size, d, and measurement number, l.

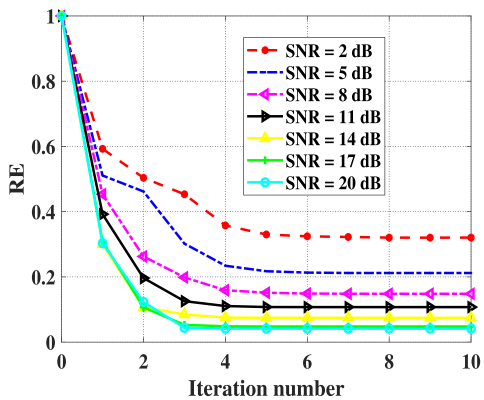


Fig. 5. The convergence of the SBSP algorithm with different SNR.

where $\mathbf{h}$ is the true CIR, $\tilde{\mathbf{h}}^i$ denotes the estimated CIR at the i-th iteration. For this purpose, we assume the actual channel $\mathbf{h}$ is known in advance. The length of the observation window is $M = 200$, and the sparse LR-UWAC in Fig. 6 is adopted for our simulation investigation. Fig. 4 plots the convergence of the SBSP algorithm for SNR $=$ 20 dB with different combinations of l and d. Regardless of the parameters chosen, we observe convergence after a few iterations with a relative error of 1%. In Fig. 5 we show the convergence of SBSP algorithm for SNR values ranging between 2 dB and 20 dB. Also here we observe convergence of SBSP algorithm after a few iterations for different SNR.

D. Non-Sparse LR-UWAC Channel Estimation

1) Maximum-Likelihood Channel Estimation: Recall we model the distribution of the ML estimator $f(\mathbf{Y};\mathbf{h})$ as Gaussian in a dense multipath,

$$f(\mathbf{Y};\mathbf{h}) = \frac{1}{\pi^N \det(\mathbf{R_Y})} e^{-\mathbf{Y}^H \mathbf{R}_\mathbf{Y}^{-1} \mathbf{Y}}, \tag{27}$$

where $\mathbf{R_Y} = E[\mathbf{Y}\mathbf{Y}^H]$ is the autocorrelation matrix. Applying the natural logarithm and multiplying by -1, we formulate the estimation problem

$$\tilde{\mathbf{h}} = \underset{\mathbf{h}}{\text{argmin}} \left(\mathbf{Y}^H \mathbf{R}_\mathbf{Y}^{-1} \mathbf{Y} + \ln(\det(\mathbf{R_Y})) \right). \tag{28}$$

Applying (6), the autocorrelation matrix $\mathbf{R_Y} \in \mathbb{C}^{M\times M}$ is obtained by

$$\begin{aligned} \mathbf{R_Y} &= E(\mathbf{Y}\mathbf{Y^H}) & (29a)\\ &= E(\mathbf{X}_D \mathbf{F}_N \mathbf{h}\mathbf{h}^H \mathbf{F}_N^H \mathbf{X}_D^H) + \sigma_w^2 \mathbf{I}_M & (29b)\\ &= \mathbf{X}_D \mathbf{F}_N \mathbf{R_h} \mathbf{F}_N^H \mathbf{X}_D^H + \sigma_w^2 \mathbf{I}_M, & (29c) \end{aligned}$$

where $\mathbf{I}_M \in \mathbb{C}^{M\times M}$ is the identity matrix, and $\mathbf{R_h} = E(\mathbf{h}\mathbf{h}^H) \in \mathbb{C}^{N\times N}$ is the autocorrelation matrix of $\mathbf{h}$. Assuming the components of $h(n)$ are statistically independent,

$$\mathbf{R_h} = \text{diag}\{\sigma_h^2(0), \sigma_h^2(1), \cdots, \sigma_h^2(N-1)\}, \tag{30}$$

where $\sigma_h^2(i) = E(|h(i)|^2)$. To solve (28), our optimization criterion depends on the magnitude of each component of the channel, and we avoid using the phase information because the channel taps of the CIR are considered to be real-valued in this study. We define

$$G = \mathbf{Y}^H \mathbf{R}_\mathbf{Y}^{-1} \mathbf{Y} + \ln(\det(\mathbf{R_Y})), \tag{31}$$

and take a partial derivative of G with respect to $h(k), k \in [0, N-1]$, to obtain

$$\xi(h(k)) = \frac{\partial G}{\partial h(k)} = \mathbf{Y}^H \frac{\partial \mathbf{R}_\mathbf{Y}^{-1}}{\partial h(k)} \mathbf{Y} + \frac{\partial \ln(\det(\mathbf{R_Y}))}{\partial h(k)}. \tag{32}$$

The derivation of (32) is given in Appendix A. The gradient descent iteration for minimizing $\xi(\mathbf{h})$ is

$$\mathbf{h}^{i+1} = \mathbf{h}^i - \mu \xi(\mathbf{h}), \tag{33}$$

where μ is the step size, and i is the iteration number, and the initial $\mathbf{h}^0$ is obtained from the coarse channel estimation in (12). The steps are repeated until

$$||\mathbf{Y} - \mathbf{X}_D \mathbf{F}_N \mathbf{h}^i||_2^2 \le \varepsilon, \tag{34}$$

where ε is the noise factor.

The complexity of each estimation step is $\mathcal{O}(M^2)$,[1] and the overall complexity of the above algorithm is $\mathcal{O}(\varsigma(NM^2))$, with M being the received signal in samples, N is the length of the LR-UWAC channel, and ς is the iteration time. Next, we discuss a relaxation method for how to reduce this complexity cost.

2) A Hybrid Non-Sparse Channel Estimation: In order to reduce the computational complexity of the above non-sparse channel estimator, we make use of a hybrid estimation model to focus the channel estimation on the non-resolvable multipath components of the LR-UWAC channel.

After the coarse channel estimation, the channel's energy within an observation window is calculated to choose the non-resolvable multipath of high energy. Since the non-sparse channel is expected to have a dense multipath, a possible solution is through least square. To this end, channel $\mathbf{h}$ is modeled according to the hybrid model [25], [26], [27],

$$\mathbf{h} = \mathbf{b}_s \odot \mathbf{h}^{LS} + \mathbf{h}^d, \tag{35}$$

where $\mathbf{b}_s \odot \mathbf{h}^{LS}$ reflects multipath components of small energy, $\mathbf{b}_s \in \{0,1\}^N$ is the channel pattern, $\mathbf{h}^d$ is the non-resolvable

[1]See (50) in the Appendix.

multipath of the LR-UWAC channel, and $\odot$ is the component-wise (Schur) product.

We obtain $\mathbf{b}_s \in \{0,1\}^N$ using an energy detector over the result of the LS, where $\mathbf{b}_s = 1$ represents the sparse component, and $\mathbf{b}_s = 0$ represents the non-resolvable component. Thus, the time delay of the non-resolvable multipath can be expressed by $\mathbf{v} \in \mathrm{index}(\mathbf{b}_s = 0)$. Let the time delay of the non-resolvable multipath be $\mathbf{v} = [v_0, v_1, \cdots, v_{p-1}]$, and let the corresponding amplitude of the non-resolvable multipath be $\mathbf{h}^d = [h^d(0), h^d(1), \cdots, h^d(p-1)]$, where p is the path number of the non-resolvable multipath for the LR-UWAC channel.

Finally, we combine the non-resolvable multipath with the small energy multipath. Formally,

$$\mathbf{h} = \mathbf{b}_s \odot \mathbf{h}^{LS} + \sum_{n=0}^{p-1} h^d(n)\delta(t - v_n). \tag{36}$$

The computational complexity of the above hybrid non-sparse algorithm is $\mathcal{O}(\varsigma(pM^2) + M\log M + wN)$, with M being the number of samples of the received signal, N is the length of the LR-UWAC channel, w is the length of an observation window for channel's energy detection, and ς is the iteration time. The LS and channel's energy detection require a computational complexity of $\mathcal{O}(M\log M)$ and $\mathcal{O}(wN)$, respectively [16]. Thus, compared to the estimator in Section IV-D.1, the relaxation presented in Section IV-D.2 reduces the computational complexity from $\mathcal{O}(M^2)$ to $\mathcal{O}(\varsigma(pM^2) + M\log M + wN)$.

V. Numerical Simulation

A. Simulation Setup

In this section, we report results of a numerical evaluation of our LR-UWAC channel estimation. For the channel model, we consider the PE propagation model [40]. While ray tracing is a common channel model, for low-frequency and long-range transmission, it is less accurate than the PE model [48]. We explore the performance for both sparse and non-sparse channels. To this end, we alter the location of the receiver and the transmitter to be either close to the sea surface or to the sea bottom. In particular, when the receiver and transmitter depths are lower than the first dominant bathymetric layer in the sound channel, the sound beams tend to scatter more [49]. In turn, this scattering can effectively reduce sea surface reflections to yield a sparse-like channel structure.

Based on the PE model, we generate 10000 CIRs for a transmitter-receiver distance of 100 km. To generate both sparse and non-sparse CIR, we place the receiver and the transmitter, uniformly determined at random between 30 m and 170 m. In each case, the depth difference between the receiver and the transmitter is less than 10 m. We randomly select three sea bottom parameters, namely sound speed, density, and attenuation from [1780 m/s, 2.10 g/cm^3, 0.53 dB/m], [1680 m/s, 1.90 g/cm^3, 0.46 dB/m], [1620 m/s, 1.80 g/cm^3, 0.39 dB/m], and [1580 m/s, 1.65 g/cm^3, 0.28 dB/m], respectively, which deviates from a hard bottom to a soft bottom. The perturbation method of scattering model [50] is applied to approximate acoustic interactions with rough sea surfaces in our simulation. We consider both zero and negative gradient sound speed profiles, with a sound speed on the sea surface generated uniformly at random distributions of 1500 m/s and 1540 m/s. An example of sparse and non-sparse channel realizations - one when the transmitter is below the sound layer and one when it is above the sound layer - are shown in Fig. 6. The simulation parameters are presented in Table II. In long range transmission, the multipaths can be extremely sparse, leaving only a single dominant path. This is captured in the CIR model when the depths of the receiver and the transmitter are much lower than the first dominant bathymetric layer in the sound speed profile. As shown in Fig. 6(a), the sparse LR-UWAC channel shows a block structure, where the significant amplitude taps of the channel are kept in a contiguous region and the remaining taps of the channel have quite small energy.

TABLE II
The Simulation Parameters

Description	Value
Carrier frequency	1150 Hz
Bandwidth	700 Hz
Subcarrier number	256
Block size	4
Continuous measurement number	2
Channel impulse response duration	340 ms
Length of discrete observation window	200
Error tolerance of the SL0 and the LS algorithm	1e-4
Covariance matrix size of the MUSIC algorithm	100
Forgetting factor of the RLS algorithm	0.99
Bernoulli parameter q of the Hybrid algorithm	1e-3
Step size of the novel ML algorithm	1e-3

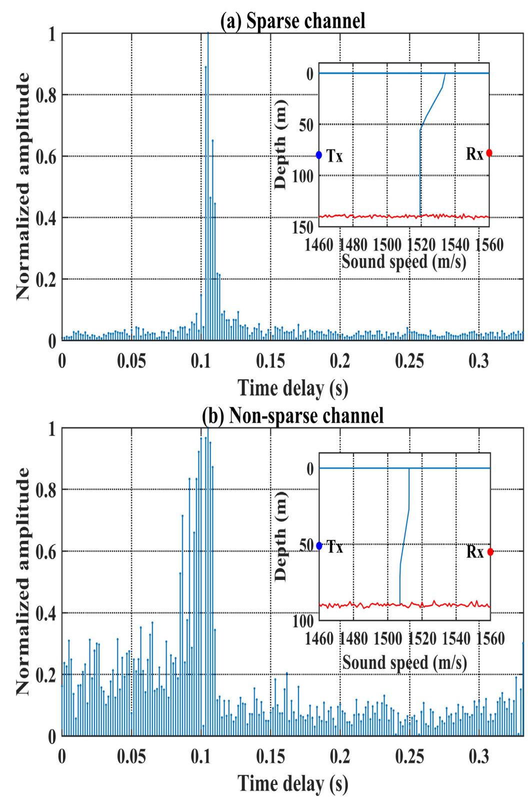


Fig. 6. LR-UWAC sparse and non-sparse channel types.

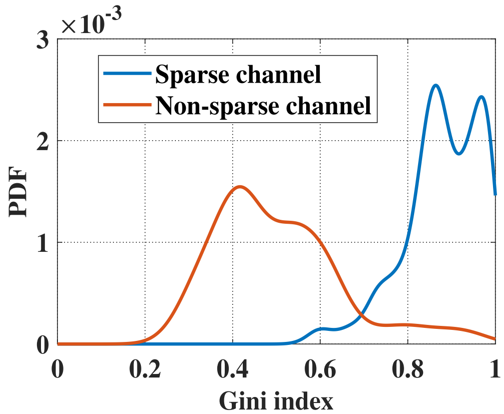


Fig. 7. The PDF of the Gini index with sparse and non-sparse LR-UWAC channels.

We compare the performance of our channel estimation with the following benchmarks. For sparse channel estimation, we use the SL0 [14], the OMP [10], the BSP [12], and the DSP [13] algorithms. The first represents a norm-type compressed sensing approach; the second is a matching pursuit-based channel estimator; and the remaining two are subspace pursuit type algorithms, which are suitable for block structure and joint sparsity channel reconstruction. We argue that all four methods are widely used in the literature. As benchmarks for non-sparse channel estimation, we consider the LS [16], the RLS [17], and the hybrid [27] channel estimators. The first two provide an optimal estimation for high SNR signals in terms of the mean square error, and the latter is a hybrid sparse/diffuse model algorithm that is suitable for complex channel reconstruction. We also consider the MUSIC algorithm [34], which provides high resolution channel estimation, and may thus be applied to both sparse and non-sparse channels.

Our simulation performance metrics are the channel-to-reconstruction error ratio (CRER) and the bit error rate (BER). The BER is calculated by the received symbols that are recovered by the linear minimum mean-square error (MMSE) equalizer [47]. The CRER is defined by

$$\text{CRER} = 10\log_{10}\frac{||\mathbf{h}||_2^2}{||\mathbf{h}-\tilde{\mathbf{h}}||_2^2}, \tag{37}$$

where $\mathbf{h}$ is the true CIR and $\tilde{\mathbf{h}}$ denotes the estimated CIR.

B. Simulation Results

We start by examining the Gini index performance as a metric to differentiate between sparse and non-sparse channels. The PDF of the Gini index for the two simulated channel types is shown in Fig. 7. Using these results, we set threshold ψ to 0.71, which yields a 93% accuracy in discriminating between sparse and non-sparse channels.

Fig. 8(a) and Fig. 8(b) show the CRER and BER results for the simulated sparse LR-UWAC channel in Fig. 6(a). We observe that the CRER converges as the SNR increases. This is because, for high SNR, channel interference becomes more dominant. The results show that the proposed SBSP estimator outperforms the benchmark estimators with an average gain of 1.1 dB and 0.05 over OMP in terms of the CRER and the BER, respectively. Compared with BSP and DSP,

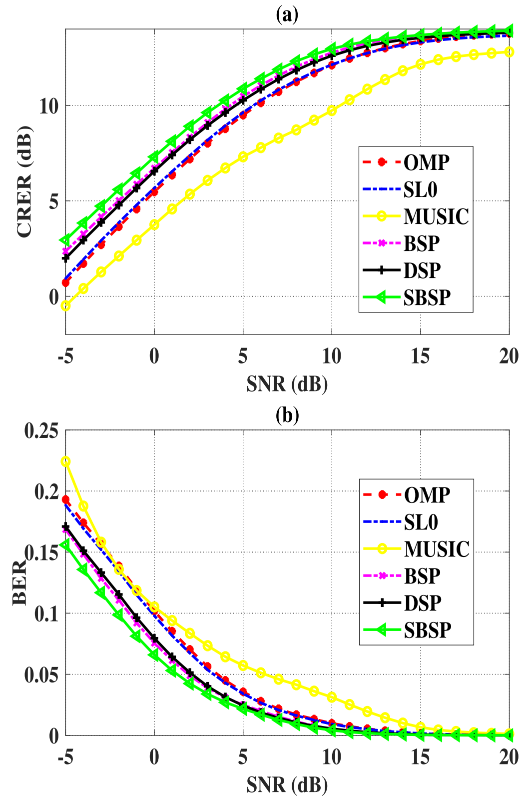


Fig. 8. The outputs of different algorithms with different SNR under the sparse LR-UWAC simulation channel. (a) CRER. (b) BER.

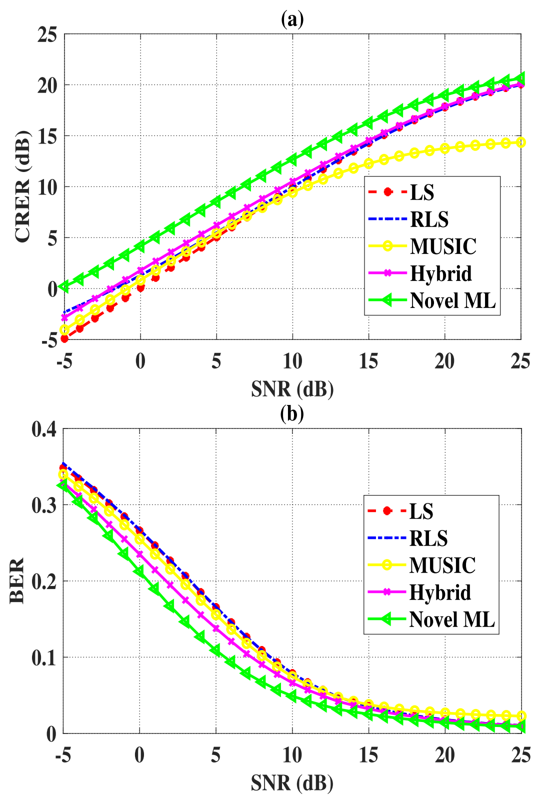


Fig. 9. The outputs of different algorithms with different SNR under the non-sparse LR-UWAC simulation channel. (a) CRER. (b) BER.

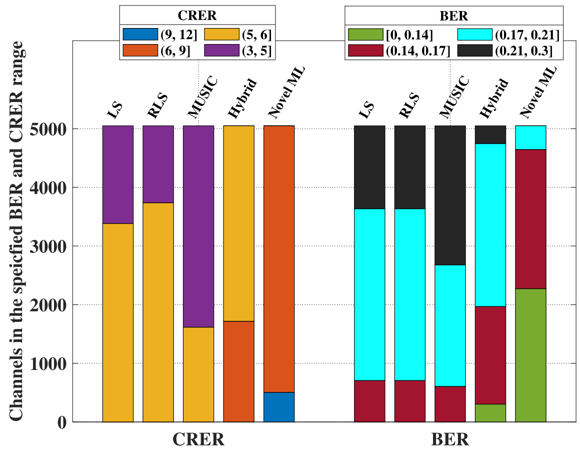


Fig. 10. The CRER and BER outputs of different algorithms with sparse LR-UWAC simulation channels.

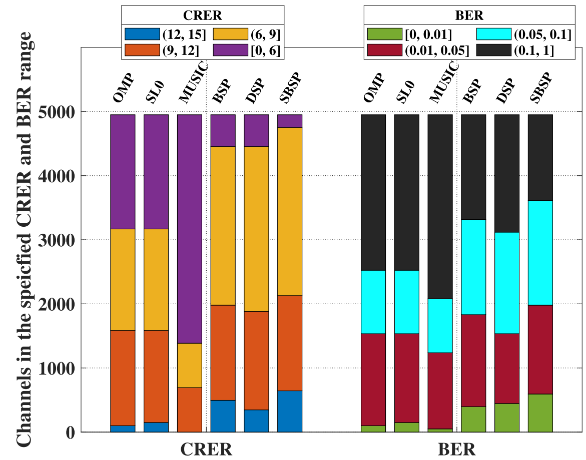


Fig. 11. The CRER and BER outputs of different algorithms with non-sparse LR-UWAC simulation channels.

we observe that the CRER and BER of OMP and SL0 are worse, due to the BSP and DSP's exploitation of the block structure and joint sparsity, respectively. The results of MUSIC fall even lower, since its requirement for the separability of paths is not met.

Next, we investigate performance under the non-sparse LR-UWAC channel in Fig. 6(b). The results in Fig. 9 show an improvement of roughly 2 dB and 0.01 in the values of the CRER and BER, respectively. In particular, the performance of MUSIC, RLS, and LS are affected by the energy concentration of the non-sparse LR-UWAC channel. The hybrid scheme offers better performance due to the existence of diffuse and sparse components in the multipath. However, the Hybrid algorithm simplifies the usage of the Rayleigh fading assumption for the diffuse component, and thus may suffer from model mismatch.

Next, we investigate performance under different LR-UWAC channels. Fig. 10 shows a histogram for the CRER and BER as obtained for different sparse LR-UWAC channels at SNR $=5$ dB. The proposed SBSP method obtains the highest CRER of 9.03 dB, versus 7.59 dB by OMP, 7.54 dB by SL0, 5.2 dB by MUSIC, 8.35 dB by BSP and 8.64 dB by the DSP algorithm. Moreover, SBSP yields the lowest BER, with a corresponding reduction of 0.020, 0.019, 0.043, 0.010, and 0.005 over the OMP, SL0, MUSIC, BSP, and DSP, respectively. We conclude that, due to the exploitation of time-invariant characteristic and the block structure in the sparse LR-UWAC channel, the proposed SBSP algorithm offers improved communication performance.

Fig. 11 shows the CRER and the BER results for different non-sparse LR-UWAC channels at SNR $=5$ dB. We observe that the proposed ML algorithm achieves better results than the benchmark estimators, with an average gain of 3.32 dB in terms of the CRER and 0.05 in terms of the BER. This is because, by modeling the received signal as Gaussian, the proposed ML algorithm can also exploit the non-resolvable multipath, thereby improving non-sparse LR-UWAC channel estimation performance.

Comparing the CRER and the BER results in Fig. 11 to those in Fig. 10, where the LR-UWAC channel structure is non-sparse, we observe an expected performance decrease due to the serious multipath interference for the LR-UWAC. For example, while in Fig. 10 at SNR $=5$ dB, more than 43% of the simulated cases achieve CRER performance that is more than or equal to 9 dB for the SBSP algorithm, in Fig. 11, that percentage decreases to 13% for the proposed ML estimator. However, comparing the performance decrease with that of the non-sparse channel estimators, we argue that our proposed ML estimator is better suited to the case of non-sparse channels.

VI. Sea Experiment

A. Experimental Setup

The above numerical investigation uses a numerical model for the acoustic channel. Especially in the case of complex channels such as the LR-UWAC, results from field trials are important to demonstrate the applicability of the proposed method. In this section, we report results from two sea experiments where no training is performed, and the threshold value of Gini index used is that obtained from the simulations. The first was conducted in March 2019 and included a low-frequency ($800-1500$ Hz) transmitter that was deployed at a depth of 20 m in Haifa Bay, Israel, where the water depth was 50 m, and included two receivers deployed at a depth of 300 m and 100 m some 100 km west of Haifa, where the water depths are roughly 1800 m. Locations of transmitter and receivers in the experiment are given in Fig. 12(a), and a full description of this experiment is available in [1]. Examples of recorded channels from this experiment are given in Fig. 1. In the following, we refer to this experiment as *Exp1*. The parameters of *Exp1* are presented in Table III.

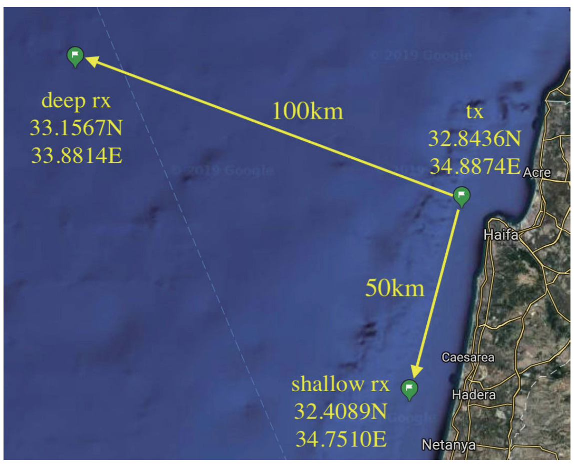
To enlarge the variation for both sparse and non-sparse channels, we performed another sea experiment, referred to here as *Exp 2*. This experiment was conducted in March 2022 and included low-frequency (1000-1500 Hz) transmissions from a 203 dB re 1μPa at 1 m omni-directional projector that was deployed in Haifa Bay, Israel, where the water depth is 60 m. During the experiment, we performed a slow start of the projector, and increased the power in steps of 5dB over a span of 1 hour. We note that this operation received

TABLE III
THE PARAMETERS OF *Exp1*

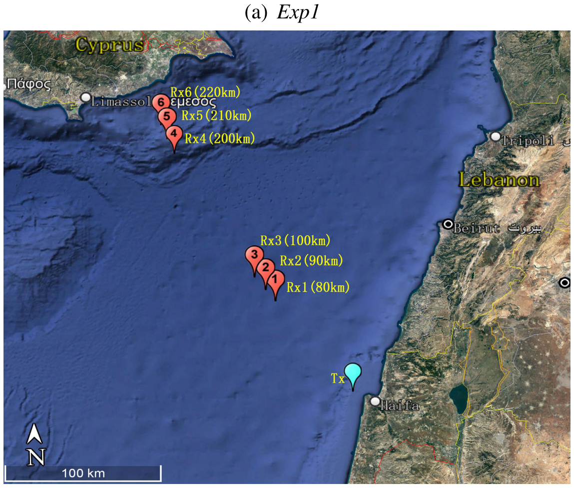
Description	Value
Carrier frequency	1200 Hz
Bandwidth	550 Hz
Subcarrier number	256
Block size	2
Continuous measurement number	2
Channel impulse response duration	300 ms
Length of discrete observation window	288
Error tolerance of the SL0 and the LS algorithm	1e-4
Covariance matrix size of the MUSIC algorithm	100
Forgetting factor of the RLS algorithm	0.99
Bernoulli parameter q of the Hybrid algorithm	1e-3
Step size of the novel ML algorithm	1e-3
Data rate	375 bps
Symbol duration	1 s
Subcarrier number	275
Number of pilot	137
Coding rate	1/2

TABLE IV
THE PARAMETERS OF *Exp2*

Description	Value
Carrier frequency	1250 Hz
Bandwidth	500 Hz
Sampling frequency	9600 Hz
Block size	2
Continuous measurement number	2
Channel impulse response duration	300 ms
Length of discrete observation window	288
Error tolerance of the SL0 and the LS algorithm	1e-4
Covariance matrix size of the MUSIC algorithm	100
Forgetting factor of the RLS algorithm	0.99
Bernoulli parameter q of the Hybrid algorithm	1e-3
Step size of the novel ML algorithm	1e-3
Data rate	125 bps
Symbol duration	1 s
Subcarrier number	250
Number of pilot	125
Coding rate	1/2



(a) *Exp1*



(b) *Exp2*

Fig. 12. Locations of the transmitter and receivers during experiments. (a) *Exp1*. (b) *Exp2*.

the authorization of the Israeli's marine protection authority. Deployment depths fluctuated during the trial between 40 m – 50 m. Several receivers were deployed at distances of 100 km, 90 km, and 80 km northwest from the transmitter, where the water depth is roughly 2000 m and the receivers were deployed at depths of 100 m and 300 m. Other receivers were deployed at distances of 220 km, 210 km, and 200 km from the transmitter, close to Cyprus, where the water depth is roughly 500 m, and the receivers were deployed at a depth of 100 m. Transmitter and receiver locations during *Exp2* are illustrated in Fig. 12(b). All locations and deployment depths were chosen by running the PE model [40] in the search for diverse channels and SNR values above 10 dB. The parameters of *Exp2* are presented in Table IV. Examples of channels recorded during *Exp2* are given in Fig. 13.

The results from *Exp1* and *Exp2* are analyzed in terms of the CRER and BER. For *Exp1*, we explored the results by using recordings of 18 CIRs in an emulation framework. That is, an OFDM signal was passed through the recorded channels and sea noise recorded during the trial was added to the result at a target SNR value. For *Exp2*, we used the same emulation approach to explore the CRER results of 144 recorded channels. However, for BER, we show results obtained by the modulated signals we transmitted - in particular, OFDM signals whose parameters are provided in Table IV.

B. Performance Evaluation

1) Exp1: We start by analyzing the success in determining the channel's type using the Gini index. For ground truth, we manually categorized the recorded channels as either 'sparse' or 'non-sparse'. Fig. 14 shows the results of the automatic categorization for *Exp1*. We observe that the threshold $\psi = 0.71$ set in the simulations can also perform well for discriminating the real channels.

Next, in Fig. 15 we show the results of the channel estimation for the sparse LR-UWAC channel shown in Fig. 1(a) for SNR values ranging between -5 dB and 20 dB. Comparing the results for all benchmarks, we note that the proposed SBSP estimator achieves the best results with an average gain of 0.95 dB in CRER and an average reduction of 0.0074 in BER. Similarly, the results in Fig. 16(a) and Fig. 16(b), show that the proposed ML algorithm can successfully recover the non-sparse LR-UWAC channel better than the benchmarks, as shown in Fig. 1(b).

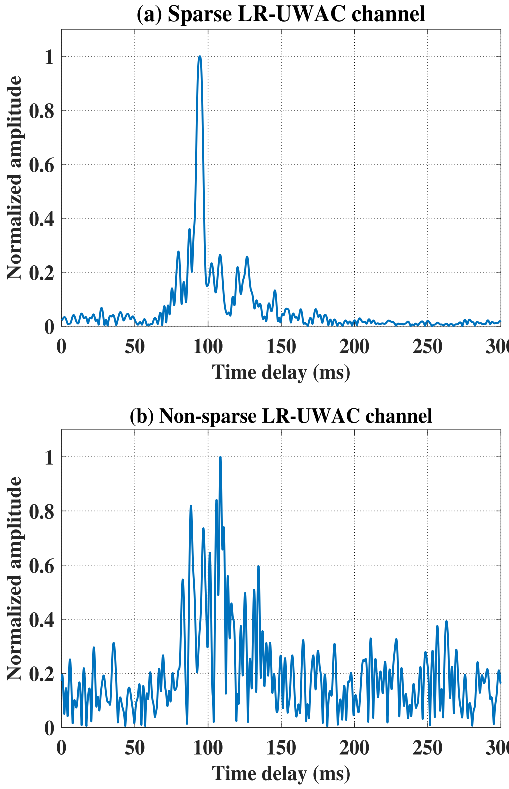


Fig. 13. Different sparse structures of the LR-UWAC channel at *Exp2*.

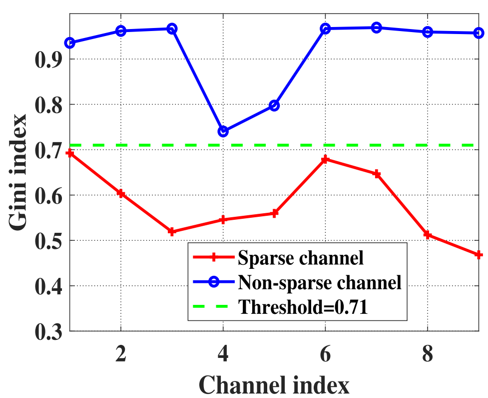


Fig. 14. The Gini index of sparse and non-sparse LR-UWAC channels at *Exp1*.

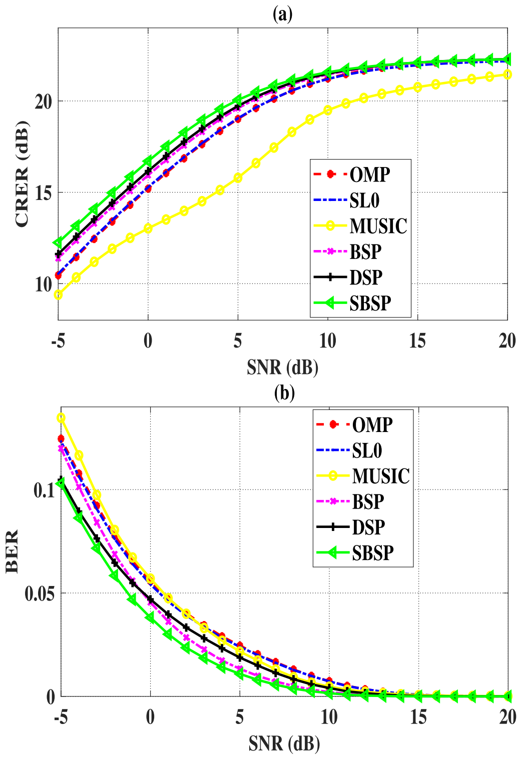


Fig. 15. The outputs of different algorithms with different SNR under the sparse LR-UWAC channel at *Exp1*. (a) CRER. (b) BER.

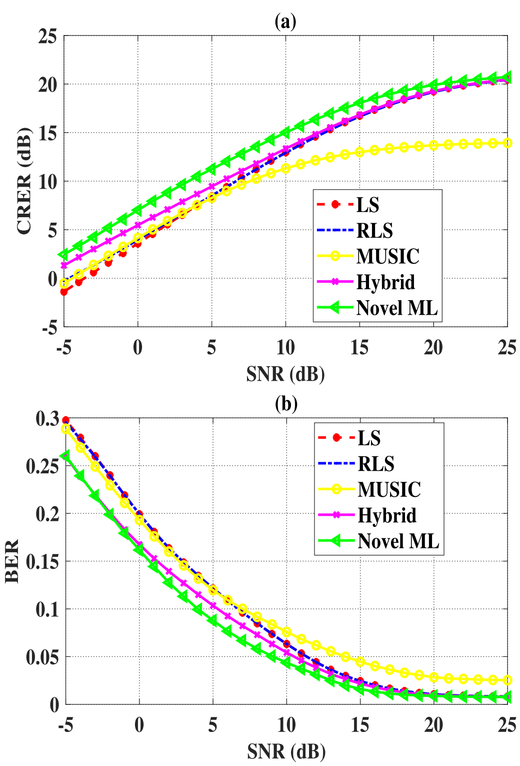


Fig. 16. The outputs of different algorithms with different SNR under the non-sparse LR-UWAC channel at *Exp1*. (a) CRER. (b) BER.

Next, we investigate the results for the 18 recorded channels. We set the SNR to 5 dB and examine the CRER and BER for the sparse LR-UWAC channels in Fig. 17, and the results for the non-sparse LR-UWAC channels in Fig. 18. We explain the results obtained by SBSP for the sparse channels by the diversity of the channel's block characteristics. Exploiting this gain, our proposed SBSP algorithm outperforms the benchmarks in terms of both CRER and BER. In contrast, the gain obtained by SBSP over the benchmarks is explained by the modeling of the received signal as Gaussian, a model that seems suitable for the measured channel.

The different performance gain observed for the different channels is explained by the various multipath distortion, which affects each method differently. In particular, we note that the gain achieved by our method increases with the ISI. We argue that this is because the received signal is more similar to Gaussian in dense multipath, which matches our proposed model.

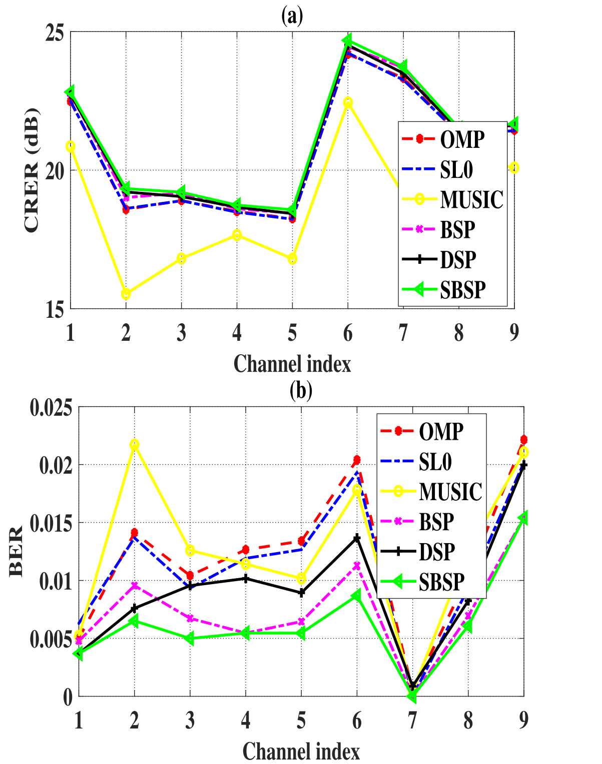


Fig. 17. The outputs of different algorithms with sparse LR-UWAC channels at *Exp1* for SNR = 5 dB. (a) CRER. (b) BER.

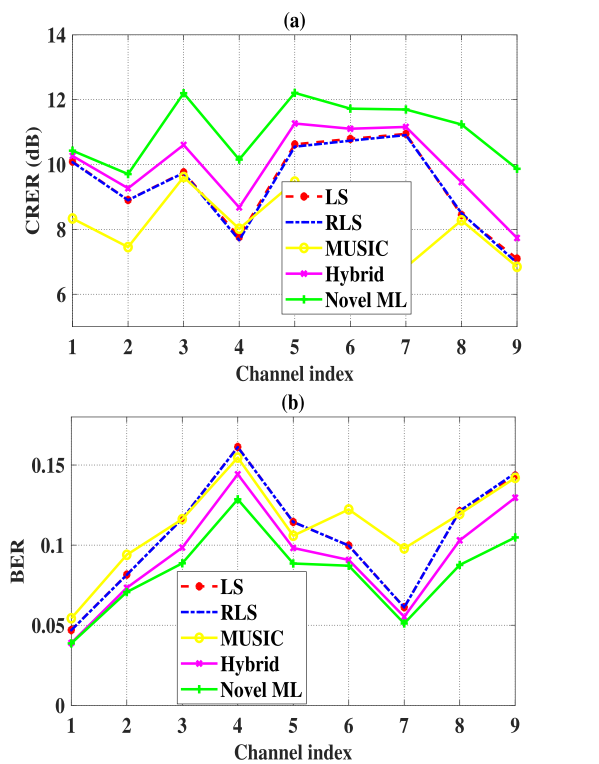


Fig. 18. The outputs of different algorithms with non-sparse LR-UWAC channels at *Exp1* for SNR = 5 dB. (a) CRER. (b) BER.

2) Exp2: As with *Exp 1*, we first examine the performance of the Gini index as a metric to separate sparse and non-sparse channels. Here, we separated 144 recorded channels to yield the PDF of the Gini index in Fig. 19. Setting threshold ψ to 0.71 as determined in the simulations, we note that 30% and 70% of the channels at *Exp2* are declared as sparse and non-sparse, respectively. Table V shows the confusion matrix for the channel's sparsity classification as verified manually. We observe that the accuracy obtained is roughly 90 % for both channel types.

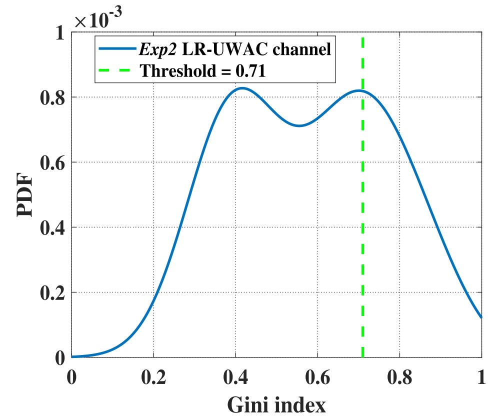


Fig. 19. The PDF of the Gini index for the *Exp2* LR-UWAC channels.

TABLE V
THE CONFUSION MATRIX FOR THE CHANNEL SPARSITY CLASSIFICATION

Confusion matrix / Threshold / Manual	Sparse	Non-sparse
Sparse	37	10
Non-sparse	5	92

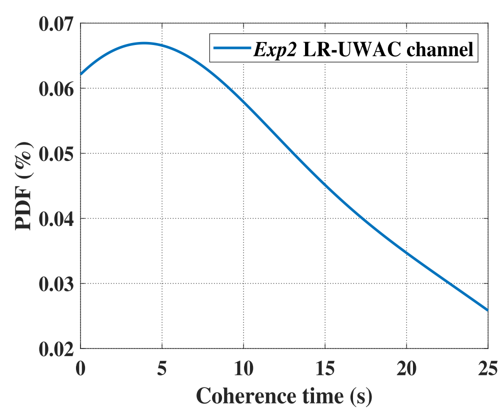


Fig. 20. The PDF of the coherence time for the *Exp2* LR-UWAC channels.

To investigate the stability of the LR-UWAC channel, Fig. 20 shows the PDF of the coherence time for the *Exp2* LR-UWAC channel. The coherence time is measured as the duration for which the channel's correlation coefficient drops below 0.9 [51]. We observe that, in most cases, the coherence time is roughly 4 seconds for the LR-UWAC channels recorded in *Exp2*.

Next, in Fig. 21 we use the above-mentioned emulation setup with an SNR of 5 dB to show the channel estimation results by the CRER metric. Here, we differentiate between the sparse and non-sparse channels. We observe that, compared to the benchmarks, the SBSP and the proposed ML estimator can successfully recover the LR-UWAC channel. Comparing the CRER results to the sparse and non-sparse channels,

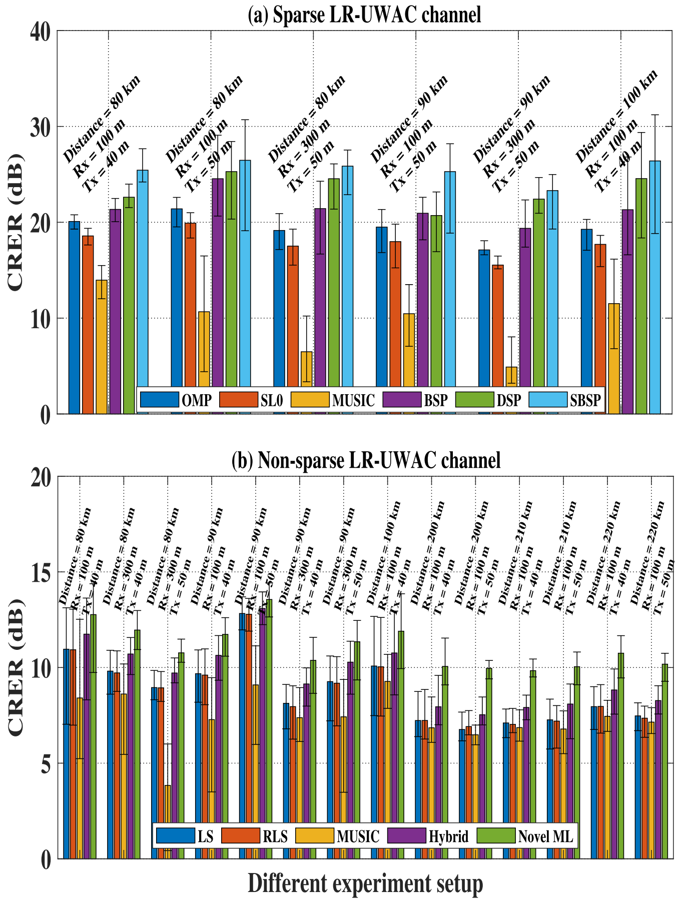


Fig. 21. The CRER outputs of different algorithms and different scenarios at *Exp2* for SNR = 5 dB. (a) Sparse LR-UWAC channels. (b) Non-sparse LR-UWAC channels.

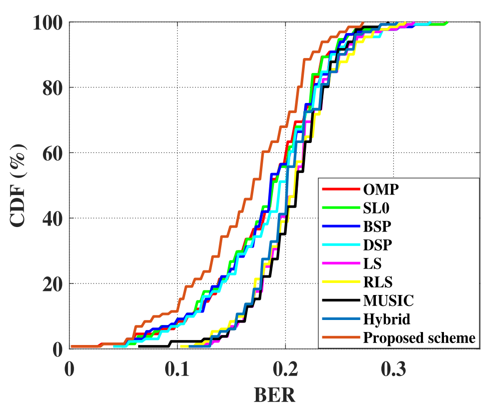


Fig. 22. The CDF of BER at *Exp2* LR-UWAC.

we observe that severe multipath propagation of non-sparse channels would introduce ISI and cause the LR-UWAC performance's degradation. Finally, in Fig. 22 we show the BER results as a cumulative distribution function (CDF) for the mixture of channels to show the full operation of our approach. That is, following the complete flow diagram in Fig. 3. We observe that the BER values of our approach are lower compared to the benchmarks. We argue that this is because the proposed approach can handle the diverse channel structure for LR-UWAC, while the benchmarks only fit the case of either a sparse or a non-sparse LR-UWAC channel.

VII. Conclusion

In this paper, we propose a robust channel estimator for LR-UWAC. A switching mechanism is first employed to differentiate between sparse and non-sparse channel structures. In the case of a sparse channel, we employ a non-trivial combination of the DSP and BSP algorithms that exploits the block structure and time-invariant characteristics of the LR-UWAC channel. For a non-sparse channel with dense multipath, we formulate an ML channel estimator for LR-UWAC that aims to separate the non-resolvable multipath. Our results from both numerical simulation and from two long-range sea experiments, including a transmission distance of 220 km from Haifa to Cyprus, demonstrate success in discriminating between sparse and non-sparse LR-UWAC channels, and an advantage over benchmark algorithms in terms of BER performance and channel estimation. Future research will include the design of a channel equalizer to combat the serious ISI in the LR-UWAC, while exploiting the channel's stability using a non-causal receiver.

Appendix A
Derivative of G

We define

$$\begin{aligned} \mathbf{F}_N &= [\mathbf{f}_0, \mathbf{f}_1, \cdots, \mathbf{f}_{N-1}], && (38a)\\ \mathbf{f}_n \in \mathbb{C}^{M\times 1} &= [\mathbf{f}_n(0), \mathbf{f}_n(1), \cdots, \mathbf{f}_n(M-1)], && (38b)\\ \mathbf{f}_n(m) &= 1/\sqrt{M}[e^{-j\frac{2\pi}{M}mn}], \\ & \quad m \in [0, M-1], n \in [0, N-1]. && (38c) \end{aligned}$$

Noting that $|X(k)|^2 = 1$, we derive the inverse of $\mathbf{R_Y}$ by

$$\begin{aligned} \mathbf{R}_{\mathbf{Y}}^{-1} &= (\mathbf{X}_D \mathbf{F}_N \mathbf{R_h} \mathbf{F}_N^H \mathbf{X}_D^H + \sigma_w^2 \mathbf{I}_M)^{-1} \\ &= (\mathbf{X}_D^H)^{-1} (\mathbf{F}_N \mathbf{R_h} \mathbf{F}_N^H + \sigma_w^2 \mathbf{I}_M)^{-1} \mathbf{X}_D^{-1}. \end{aligned} \quad (39)$$

A. Derive the First Term in (32)

Let

$$\mathbf{R_{Fh}} = \mathbf{F}_N \mathbf{R_h} \mathbf{F}_N^H + \sigma_w^2 \mathbf{I}_M, \quad (40)$$

the derivative of the inverse of $\mathbf{R_{Fh}}$ with respect to $h(k)$ is given by [52]

$$\begin{aligned} \frac{\partial \mathbf{R}_{\mathbf{Fh}}^{-1}}{\partial h(k)} &= -\mathbf{R}_{\mathbf{Fh}}^{-1} \frac{\partial \mathbf{R_{Fh}}}{\partial h(k)} \mathbf{R}_{\mathbf{Fh}}^{-1} \\ &= -2h(k) \mathbf{R}_{\mathbf{Fh}}^{-1} \mathbf{F}_N \mathbf{\Phi}(k) \mathbf{F}_N^H \mathbf{R}_{\mathbf{Fh}}^{-1} \\ &= -2h(k) \mathbf{R}_{\mathbf{Fh}}^{-1} \mathbf{f}_k \mathbf{f}_k^H \mathbf{R}_{\mathbf{Fh}}^{-1}, \end{aligned} \quad (41)$$

where

$$\begin{aligned} \mathbf{\Gamma}(k) \in \mathbb{C}^{1\times N} &= [0, \cdots, 0, 1, 0, \cdots, 0], && (42a)\\ \mathbf{\Phi}(k) &= \mathrm{diag}\{\mathbf{\Gamma}(k)\}. && (42b) \end{aligned}$$

Since $\mathbf{F}_N^H \mathbf{F}_N = \mathbf{I}_N$, thus

$$\mathbf{f}_i^H \mathbf{f}_k = \begin{cases} 1, i = k \\ 0, i \neq k, \end{cases} \; i \in [0, N-1], \quad (43)$$

and (41) can be rewritten as

$$\begin{aligned}\frac{\partial \mathbf{R}_{\mathbf{Fh}}^{-1}}{\partial h(k)} &= -h(k)(\mathbf{F}_N\mathbf{R}_{\mathbf{h}}\mathbf{F}_N^H+\sigma_w^2\mathbf{I}_M)^{-1}\mathbf{f}_k\mathbf{f}_k^H\mathbf{R}_{\mathbf{Fh}}^{-1}\\ &= -2h(k)((\frac{1}{\sigma_h^2(k)}\mathbf{F}_N\mathbf{\Phi}(k)\mathbf{f}_k^H)^{-1}\\ &\quad +\sigma_w^2(\mathbf{f}_k\mathbf{f}_k^H)^{-1})^{-1}\mathbf{R}_{\mathbf{Fh}}^{-1}\\ &= \frac{-2h(k)}{\sigma_h^2(k)+\sigma_w^2}\mathbf{f}_k\mathbf{f}_k^H(\mathbf{F}_N\mathbf{R}_{\mathbf{h}}\mathbf{F}_N^H+\sigma_w^2\mathbf{I}_M)^{-1}\\ &= \frac{-2h(k)}{\sigma_h^2(k)+\sigma_w^2}((\frac{1}{\sigma_h^2(k)}\mathbf{f}_k\mathbf{\Phi}(k)\mathbf{F}_N^H)^{-1}\\ &\quad +\sigma_w^2(\mathbf{f}_k\mathbf{f}_k^H)^{-1})^{-1}\\ &= \frac{-2h(k)}{(\sigma_h^2(k)+\sigma_w^2)^2}\mathbf{f}_k\mathbf{f}_k^H. \end{aligned} \tag{44}$$

Hence, the first term in (32) can be expressed by

$$\begin{aligned}\mathbf{Y}^H\frac{\partial \mathbf{R}_{\mathbf{Y}}^{-1}}{\partial h(k)}\mathbf{Y} &= \mathbf{Y}^H(\mathbf{X}_D^H)^{-1}\frac{\partial \mathbf{R}_{\mathbf{Fh}}^{-1}}{\partial h(k)}\mathbf{X}_D^{-1}\mathbf{Y}\\ &= \frac{-2h(k)}{(\sigma_h^2(k)+\sigma_w^2)^2}(\mathbf{X}_D^{-1}\mathbf{Y})^H\mathbf{f}_k\mathbf{f}_k^H\mathbf{X}_D^{-1}\mathbf{Y}\\ &= \frac{-2h(k)}{(\sigma_h^2(k)+\sigma_w^2)^2}(\mathbf{f}_k^H\mathbf{X}_D^{-1}\mathbf{Y})^H\mathbf{f}_k^H\mathbf{X}_D^{-1}\mathbf{Y}. \end{aligned} \tag{45}$$

B. Derive the Last Term in (32)

The determinant of $\mathbf{R}_{\mathbf{Y}}$ is calculated by

$$\begin{aligned}\det(\mathbf{R}_{\mathbf{Y}}) &= \det(\mathbf{X}_D\mathbf{F}_N\mathbf{R}_{\mathbf{h}}\mathbf{F}_N^H\mathbf{X}_D^H+\sigma_w^2\mathbf{I}_M)\\ &= \det(\mathbf{X}_D)\det(\mathbf{F}_N\mathbf{R}_{\mathbf{h}}\mathbf{F}_N^H+\sigma_w^2\mathbf{I}_M)\det(\mathbf{X}_D^H)\\ &= \det(\mathbf{F}_N\mathbf{R}_{\mathbf{h}}\mathbf{F}_N^H+\sigma_w^2\mathbf{I}_M)\\ &= \sigma_w^{2M}\det(\sigma_w^{-2}\mathbf{F}_N\mathbf{R}_{\mathbf{h}}\mathbf{F}_N^H+\mathbf{I}_M). \end{aligned} \tag{46}$$

According to the Weinstein–Aronszajn identity theorem [53], and since $\mathbf{F}_N^H\mathbf{F}_N=\mathbf{I}_N$, $\det(\mathbf{R}_{\mathbf{Y}})$ can equivalently be

$$\begin{aligned}\det(\mathbf{R}_{\mathbf{Y}}) &= \sigma_w^{2M}\det(\sigma_w^{-2}\mathbf{R}_{\mathbf{h}}\mathbf{F}_N^H\mathbf{F}_N+\mathbf{I}_N)\\ &= \sigma_w^{2M}\det(\sigma_w^{-2}\mathbf{R}_{\mathbf{h}}+\mathbf{I}_N)\\ &= \sigma_w^{2M}\prod_{n=0}^{N-1}(\sigma_w^{-2}\sigma_h^2(n)+1). \end{aligned} \tag{47}$$

Thus,

$$\ln(\det(\mathbf{R}_{\mathbf{Y}})) = M\ln(\sigma_w^2)+\sum_{n=0}^{N-1}\ln(\sigma_w^{-2}\sigma_h^2(n)+1). \tag{48}$$

Then, for the last term in (32), we get

$$\frac{\partial \ln(\det(\mathbf{R}_{\mathbf{Y}}))}{\partial h(k)} = \frac{2h(k)}{\sigma_h^2(k)+\sigma_w^2}. \tag{49}$$

Finally, by summing up the above two terms, the derivative of G becomes

$$\begin{aligned}\frac{\partial G}{\partial h(k)} &= \frac{-2h(k)}{(\sigma_h^2(k)+\sigma_w^2)^2}(\mathbf{f}_k^H\mathbf{X}_D^{-1}\mathbf{Y})^H\mathbf{f}_k^H\mathbf{X}_D^{-1}\mathbf{Y}\\ &\quad +\frac{2h(k)}{\sigma_h^2(k)+\sigma_w^2}\\ &= \frac{-2h(k)}{(|h(k)|^2+\sigma_w^2)^2}(\mathbf{f}_k^H\mathbf{X}_D^{-1}\mathbf{Y})^H\mathbf{f}_k^H\mathbf{X}_D^{-1}\mathbf{Y}\\ &\quad +\frac{2h(k)}{|h(k)|^2+\sigma_w^2}. \end{aligned} \tag{50}$$

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